

Towards Zero-Shot Document Image Classification

Artigo WPOS

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Novembro de 2025

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Introdução

Contexto - Documentos e Compliance

- Ambiente Bancário
- Classificação de documentos
- Exemplos:



MINISTÉRIO DA FAZENDA		IMPOSTO SOBRE A RENDA - PESSOA FÍSICA	
SECRETARIA DA RECEITA FEDERAL DO BRASIL		EXERCÍCIO 2017 ANO-CALENDÁRIO 2018	
RECIBO DE ENTREGA DA DECLARAÇÃO DE AJUSTE ANUAL - OPÇÃO PELO DESCONTO SIMPLIFICADO DECLARAÇÃO ORIGINAL			
IDENTIFICAÇÃO DO DECLARANTE			
CPF do declarante 039.056-378-08		Nome do declarante ERIKA TOMAZELLA	
Telefone (11) 43791221			
Endereço RUA RUA JUSTINO ALVES BATISTA		Número 89	Complemento AP 64 BL P
Bairro/Distrito VILA YOLANDA	CEP 08126-130	Município OSASCO	UF SP
TOTAL RENDIMENTOS TRIBUTÁVEIS			(Valores em Reais) 62.260,42
IMPOSTO DEVIDO			3.562,66
IMPOSTO A RESTITUIR			781,16

Evolução dos Documentos



Zero-Shot Learning

Permite que o modelo reconheça elementos de classes nunca vistas no treinamento

Contribuições

① Novo dataset LA-CDIP

- Classificação ZSL
- Derivado do RVL-CDIP

② Abordagem de Visual Document Matching (VDM)

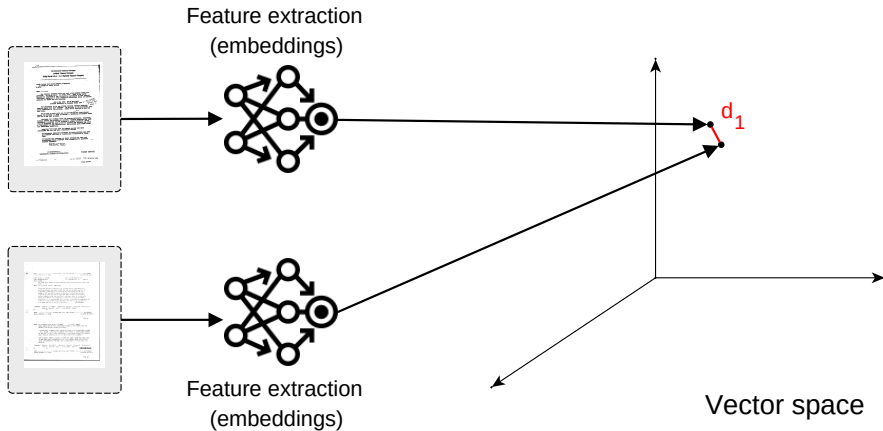
- Similaridade de documentos
- Metric Learning
- Generalização Zero-Shot

③ Avaliação sistemática

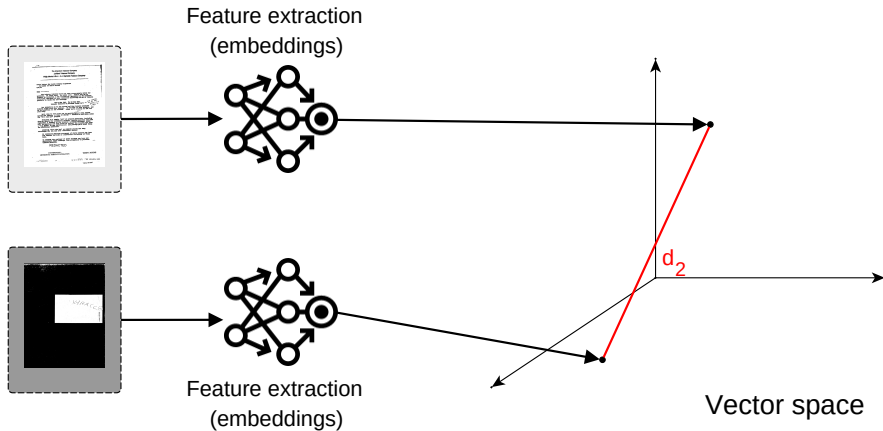
- Benchmark extensivo

Metodologia

Visual Document Matching



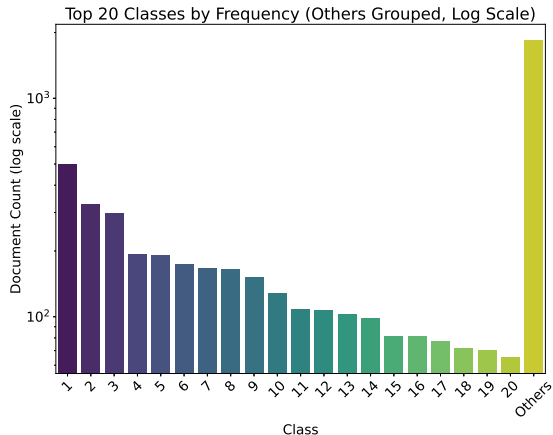
Visual Document Matching



Resultados

Composição:

- 4.993 documentos
- 144 classes diferentes
- Min: 2 documentos/classe
- Max: 497 documentos/classe
- Mediana: 13 documentos/classe



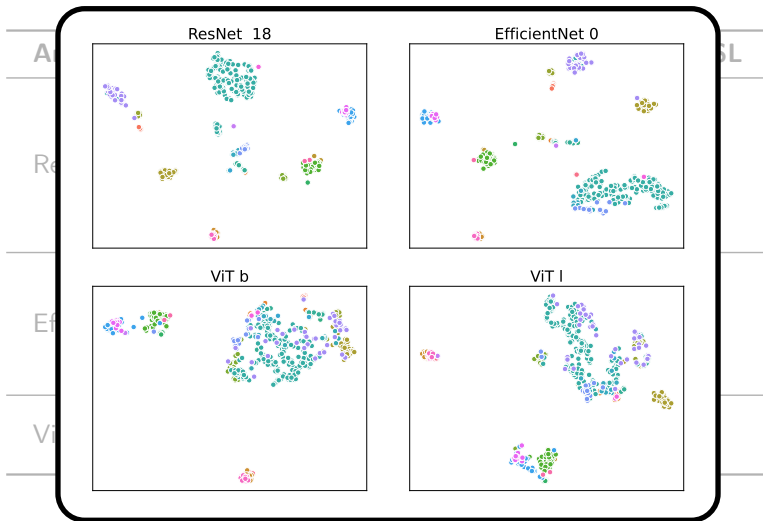
Resultados - Equal Error Rate (%)

Architecture	Edition	Params	ZSL	Test ZSL
VGG	11	129M	7.47	14.24
	13	129M	7.03	9.30
	16	134M	8.29	14.74
	19	139M	7.30	17.08
ResNet	18	11M	5.03	4.98
	34	21M	4.32	4.13
	50	23M	6.90	10.34
	101	42M	8.20	11.31
	152	58M	9.44	12.70
MobileNetV3	Small	1M	7.98	12.74
	Large	4M	8.16	8.45
EfficientNet	0	4M	4.41	6.02
	1	6M	<u>3.93</u>	8.88
	2	7M	5.73	7.29
	3	10M	5.65	7.37
ViT	Base	87M	12.43	19.72
	Large	305M	13.16	19.88

Resultados - Equal Error Rate (%)

Architecture	Edition	Params	ZSL	Test ZSL
ResNet	18	11M	5.03	4.98
	34	21M	4.32	<u>4.13</u>
	50	23M	6.90	10.34
	101	42M	8.20	11.31
	152	58M	9.44	12.70
EfficientNet	0	4M	4.41	6.02
	1	6M	<u>3.93</u>	8.88
	2	7M	5.73	7.29
	3	10M	5.65	7.37
ViT	Base	87M	12.43	19.72
	Large	305M	13.16	19.88

Resultados - Visualização TSNE



Conclusão

① LA-CDIP Dataset

Categorizado por layout visual para Zero-Shot Learning

② Metodologia com VDM

Enforçando comparação de documentos como forma de classificação

③ Benchmarking Sistemático

Comparação entre modelos visuais

Visual Document Matching for Zero-Shot Document Classification

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Abstract. Accurate document identification is crucial for ensuring compliance and maintaining consistency across various applications. Consequently, document classification has been extensively studied. However, the dynamic nature of documents often renders single classification inefficient, necessitating frequent model retraining. To mitigate this one-time maintenance, comparing incoming documents against known references is a more suitable approach. To address this one-shot classification problem, we introduce Visual Document Matching (VDM) by leveraging a unified whether two documents share an identical visual layout structure, a particularly effective approach in scenarios where training classes do not align with those encountered during inference. Despite its significant potential, the Zero-Shot Learning (ZSL) approach remains largely underexplored in document layout understanding. To support our study, we introduce Layout-Aware Complex Document Information Processing (LA-CDIP), a dataset comprising 1380 documents across 141 classes. We augmented this dataset from the RVL-CDIP database to regularize visual structure over semantic information. Our approach is benchmarked using a Siamese network within a contrastive learning framework across multiple hardware architectures, including ResNet, EfficientNet, and Vision Transformer (ViT). In unimodal scenario, our method achieves an Equal Error Rate (EER) below 1% in 1-vs-1 verification with cross-validation. Furthermore, our VDM approach outperforms English Large Language Models (LLMs) and vision GPT-4o. These findings highlight the superiority of specialized VDM techniques over general-purpose multimodal models, demonstrating high accuracy with significantly fewer parameters, making our approach more practical for real-world applications.

Keywords: Zero-shot learning · Metric Learning · Document Understanding · Document Image Classification · Visual Document Matching

1 Introduction

Documents are central to many organisations, acting as official records for contracts, identities, and evidence, and playing a crucial role in legal, financial, and

Towards Zero-Shot Document Image Classification

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Abstract—Classification is a fundamental tool to automate the process of categorizing documents in many real-world applications, such as information management, financial document processing, healthcare records management, news categorization, fraud detection, regulatory compliance, and many others. Because of this broad spectrum of applications, document classification is of paramount importance for various companies. However, documents often change in terms of format and their visual patterns, which may impede a single classification model. Moreover, model maintenance and retraining also demands important efforts, consuming computational resources and demanding new data. Therefore, techniques capable of classifying documents by simply observing new data, without necessarily requiring retraining the classifier, are of immense importance for a wide variety of applications. In this context, Zero-Shot Learning (ZSL) is especially suitable for document classification because it handles diverse and ever-changing document content. In this work, we tackle the gap involving Zero-Shot Document Image Classification (ZS-DIC), where we classify documents that have not been seen by the model during training. To achieve this, we built Layout-Aware Complex Document Information Processing (LA-CDIP), a dataset tailored for this problem. LA-CDIP prioritizes structural consistency, allowing models to identify documents correctly under a ZSL scenario. To benchmark this dataset, we developed a series of Siamese Neural Networks (SNN) based on a variety of computer vision neural architectures, such as ResNet, EfficientNet, ViT and others. As a result, the proposed ZSL-based method achieves Equal Error Rate (EER) under 1%. The code of the proposed method is available at <https://github.com/UnB/BrBibData-zsl>.

I. INTRODUCTION

Document understanding, vital for businesses, automates information extraction and interpretation from documents [1]. The prominence of this field within the business ecosystem stems from the universal necessity for companies to manage documents. Therefore, an automated pipeline for document processing is widely adopted and often essential. For instance, companies that process personal documents, such as identification documents, frequently encounter low-quality images. Nevertheless, an automated pipeline remains more efficient in both cost and time than a manual verification process. Consequently, most companies find machine learning a viable technology for automation, as deterministic methods may struggle to account for the high variability of document quality.

In this context, the prevalent approach to address this problem involves traditional classification methods, often utilizing Deep Learning (DL). The problem has been widely studied in the field of document understanding as a single classification task, mainly supported by the Ryerson Vision Lab Complex Document Information Processing (RVL-CDIP) 978-0-915-8951-6/25/\$31.00 ©2025 IEEE

dataset [2]. This dataset comprises 400,000 documents divided into 16 classes, based on their purpose, such as letters and emails. RVL-CDIP has been a subject of extensive research for traditional classification, with some works achieving over 97% accuracy [3], [4] in specific scenarios. However, existing documents frequently undergo changes in form and structure, and new document types are continuously generated. In such dynamic environments, new data must be accepted to retrain the classification model.

To overcome the limitations imposed by the need for labeled training data, ZSL emerges as the most appropriate paradigm. This approach defines a task where the classes in the test set are disjoint from those in the training set, thus forcing the model to generalize effectively to novel categories [5], [6]. Despite being a considerably under-researched area within document analysis, ZS-DIC provides valuable contributions.

Although RVL-CDIP is the most widely used dataset for document classification, other datasets are available. DocVQA [7] focuses on Visual Question Answering (VQA), DocLayoutNet [8] is specialized in layout segmentation, and CORD-90, SROIE [10], FUNSD [11], and NFD [12] were created for information extraction. While these datasets advance their respective fields, their label characteristics prevent their effective use in ZS-DIC.

To address these challenges, we introduce LA-CDIP, a dataset designed for the ZS-DIC challenge. LA-CDIP is a document-image dataset that re-poseses the images from RVL-CDIP under a new classification scheme. Unlike RVL-CDIP, which classifies documents by their purpose (e.g., emails, letters), our dataset arranges them by their visual structure, grouping together documents that share a similar layout. This organization enables ZS-DIC by ensuring that each class exhibits high consistency in both visual and textual patterns.

We propose a Visual Document Matching (VDM) framework for this dataset. This framework leverages Contrastive Learning (CL) [13] to generate a similarity score between two documents. A predefined similarity threshold is used to determine if documents belong to the same class. In a practical application, by providing a reference for each class, the framework can be deployed in two modes: as a binary classifier for verification (comparing each incoming document with a single class reference) or as a multiclass classifier for identification (comparing each incoming document with all available references to select the most similar one).

The remainder of this paper is structured as follows. Section II details our LA-CDIP dataset. Section III describes the proposed Siamese network modeling. Sections IV and Sec-

Obrigado!

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